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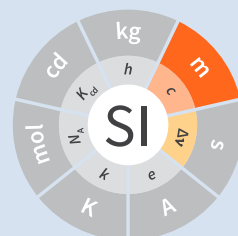
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Annexe 2
Annex 2

Recommended Values of Standard Frequencies for Applications Including the Practical Realization of the Metre and Secondary Representations of the Definition of the Second: Rubidium ()

CCL-CCTF Frequency Standards Working Group

May 2016
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**Recommended Values of Standard
Frequencies for Applications Including
the Practical Realization of the Metre
and Secondary Representations of the
Definition of the Second: Rubidium ()**

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1. Recommended value [1] of the frequency:

$$f(^{87}\text{Rb}_{d/f \text{ crossover}}) = 384\,227\,981.9 \text{ MHz}$$

equivalent to

$$\lambda(^{87}\text{Rb}_{d/f \text{ crossover}}) = 780\,246\,291.6 \text{ fm},$$

with an estimated relative standard uncertainty of 5×10^{-10} applies to the radiation of a tunable Extended Cavity Diode Laser, stabilized to the d/f saturated absorption line in a well characterized magnetically shielded rubidium absorption cell by the third-harmonic technique,

- cell temperature $(22.5 \pm 1) ^\circ\text{C}$
- frequency modulation width, peak-to-peak, $(5 \pm 1) \text{ MHz}$;
- saturating beam intensity of $(120 \pm 20) \text{ mWcm}^{-2}$

2. Source data

Adopted value $f = 384\,227\,981.9$ MHz
 $u_c/y = 5 \times 10^{-10}$

calculated from

$f(^{87}\text{Rb}_{\text{d/f crossover}})/\text{kHz}$	u_c/y	source data
384 227 981 877.3 (5.5)	1.5×10^{-11}	[2]
384 227 981 867.8 (165)	4.3×10^{-10}	[3]

as a weighted mean. The fractional uncertainty was evaluated to be approximately 4.3×10^{-10} for a 0.01 s averaging time [3]. For the recommendation this uncertainty was rounded to 5×10^{-10} .

^{87}Rb atom, crossover transition between the d and f hyperfine components of the saturated absorption at 780 nm (D2 transition)

References

- [1] CIPM Recommendation 2 (CI-2015): Updates to the list of standard frequencies
<http://www.bipm.org/jsp/en/CIPMRecommendations.jsp>
- [2] J. Ye, S. Swartz, P. Jungner, and J. L. Hall, “Hyperfine structure and absolute frequency of the ^{87}Rb $5\text{P}_{3/2}$ state”, *Opt. Lett.* **21**, 1280 (1996).
- [3] Y. Bitou, K. Sasaki, H. Inaba, F.-L. Hong, A. Onae, “Rubidium-stabilized diode laser for high-precision interferometer”, *Opt. Eng.* **43**, 900 (2004).